To try and improve upon the baseline precision scores of the dot product, cosine similarity, and TF-IDF cosine similarity baseline, I implemented a two-stage recommendation system. First, TF-IDF cosine similarity was used to gather 20 of the most similar candidate songs for each query song, where songs by the same artist for the query were removed before selecting the 20 songs. I then used the original word count vectors and standard cosine similarity to rerank these 20 candidates, selecting the final 3 recommendations. This produced a precision of ~21.43%, compared to the highest baseline precision from TF-IDF of ~21.07%. Although this was a tiny improvement, it was an improvement regardless. The reasoning for using this strategy was that we know that TF-IDF does a decent job at identifying songs based on specific words, while cosine similarity can then match up the query song to other songs simply based on the word count and patterns of this smaller candidate set. This, in theory, could combine the strengths of having a smaller set of candidates to comb through that has been reduced to songs that are generally more similar to the query song, allowing a more basic cosine similarity to match it up based on word count and word patterns. The current baseline models work decent on their own, and combining two of them does work to some extent, as I did find minimal improvement, but improvement nonetheless. Although not attempted in this lab, an idea I had was that it could be beneficial to try and combine TF-IDF with a sentiment analysis model, as that would be able to obtain the “sentiment” of a song, and thereby attach a positive/negative score to a song, adding another feature that could complement the model in deciding on a song that matches the genre.